

NumPy and Matplotlib

EEC 3150

Trevecca Nazarene University

1 RNG

- Random Walks
- RNG on Different Domains

2 Monte Carlo Simulation

RNG

Random Walks

Random Walks:

A random walk is a sequence $x_0, x_1, x_2, \dots, x_{N-1}$ of randomly generated steps through some space:

It is usually formulated

$$x_{i+1} = x_i + r_i \quad (1)$$

where $i = 0, \dots, N - 1$, x_0 is a given initial state, and the r_i 's are randomly generated from some distribution

Random Walks

Create some random walks:

1D +1 or -1

1D uniform(0,1)

1D uniform(-1,1)

1D normal distribution

2D normal

2D up-down/left-right (with bias)

RNG on Different Domains

RNG on Different Domains:

We can use NumPy's RNG tools to randomly generate points inside an arbitrary domain:

We can uniformly sample different domains by sampling along the “dimensions” of the domain

RNG on Different Domains

Uniformly sample some domains:

Circle perimeter

Rectangle

Disk area

Sphere

Monte Carlo Simulation

Monte Carlo Simulation

Monte Carlo Simulation:

Monte Carlo simulation is a computational technique used to model and simulate complex systems or processes using random sampling and statistical methods

You can use it to:

- Evaluate complex integrals over complex domains

- Evaluate statistical measures on complex distributions

- Generate complex distributions

Technique:

- Generate LARGE volume of random samples on some distribution

- Perform various statistical analyses of the result

Monte Carlo Simulation

Estimate π :

Steps:

1. Uniformly sample N_{sq} points in the unit square
2. Inscribe the unit circle in the unit square
3. Count the number N_{cl} points in the unit circle
4. For large N_{sq} , $\frac{N_{cl}}{N_{sq}} \approx \frac{A_{cl}}{A_{sq}}$
5. $\frac{A_{cl}}{A_{sq}} = \frac{\pi r^2}{(2r)^2} = \frac{\pi}{4} \approx \frac{N_{cl}}{N_{sq}}$
6. $\pi \approx 4 \frac{N_{cl}}{N_{sq}}$

Some notes:

More accurate approximations for large N_{sq}

MC has a slow rate of convergence, i.e., it takes MANY samples to get accurate results (there are more efficient means of estimating π)